MongoDB

MongoDB is a cross-platform, document oriented database that provides high performance, high-availability and easy scalability. MongoDB works on concept of collection and document

Database:-

Database is physical container for collections. Each database gets its own set of files on the file system. A single MongoDB server typically has multiple interface., database.

Collection:-

Collection is a group of MongoDB document. It is an eqivalent to RDBMS table. A collection exists within a single database. Collections do not enforce a schema document within a collection can have diffrent fields.

Document:-

A document is a set of key-value pair. Document have dynamic schema. Dynamic schema means that document in the same collection do not need to have the same set of fields and structure and common fields in a collection document may hold diffrent types of data.

Advantages of MongoDB over RDBMS :-

- Schema less :- MongoDB is a document database in which one collection holds diffrent documents Number of fields, content and size of the document can differ from one document to another.
- Structure of a single object is a clear
- No complex join
- Deep query-ability. MongoDB supports dynamic queries on document using a document-based query language that nearly as powerfull as SQL
- Tuning
- Conversion/

Why use MongoDB :

- Document Oriented Storaye.
- Index on any attribute
- Replication and availability
- Auto-Sharding
- Rich Queries
- Fast in-place updates
- Proffesional support by MongoDB

Where to use MongoDB :

① MongoDB → Big Data
Content Management and Delivery
Mobile and Social Infrastructure
User Data Management
Data hub

Start MongoDB

sudo service mongodb start

Stop MongoDB

Sudo service mongodb stop

Restart MongoDB

Sudo service mongodb restart

MongoDB - Data Modelling.

Data in MongoDB has flexible schema documents in the same collection. They do not need to have the same set of fields or structure common fields in the collections document may hold diffrent types of data

Data Model Design:-

MongoDB provides 2 types of data models.

① Embedded Data Model

In this model, you can have all the related data in a single document. It is also known as de-normalized data model.

Example:-

```
{
   _id:,
   Emp_ID: "10025AE336"
   Personal_details: {
         First_Name : "Radhika",
```